

PORTLAND CASHMORE, Australia

WMO: 948280

Lat: 38.3147S

Lon: 141.4706E

Elev: 82

StdP: 100.35

Time Zone: 10.00 (AUS)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.2	4.4	0.5	3.9	9.1	1.4	4.2	8.4	14.2	9.9	12.9	10.1	2.0	0	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.9	30.7	18.2	27.6	17.5	24.7	16.9	19.7	26.2	18.7	24.7	17.8	22.7	5.8	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.9	13.0	20.9	16.9	12.2	20.0	16.1	11.5	19.1	56.5	26.3	53.4	24.8	50.6	22.8	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.1	10.8	9.6	1.0	37.0	0.7	2.5	0.5	38.9	0.1	40.4	-0.3	41.8	-0.8	43.6		
			0.7	22.4	0.7	1.8	0.2	23.7	-0.2	24.7	-0.6	25.7	-1.0	27.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.3	16.9	17.4	16.1	13.9	12.0	10.2	9.8	10.2	11.3	12.4	14.0	15.1
	DBStd	3.69	3.42	3.24	3.06	2.54	2.15	1.72	1.64	1.80	2.21	2.82	3.09	3.19
	HDD10.0	78	0	0	0	1	5	17	24	19	8	4	1	0
	HDD18.3	1946	70	52	87	136	197	243	265	252	211	185	137	112
	CDD10.0	1270	215	208	188	118	67	24	17	24	47	80	122	159
	CDD18.3	98	27	26	17	3	1	0	0	0	1	3	8	13
	CDH23.3	914	259	213	147	32	2	0	0	0	2	37	87	134
	CDH26.7	358	116	87	56	8	0	0	0	0	9	25	56	
Wind	WSAvg	4.7	4.6	4.5	4.4	4.0	4.4	5.1	5.1	5.3	5.1	4.9	4.9	
Precipitation	PrecAvg	811	34	26	47	55	73	112	105	102	89	64	52	48
	PrecMax	998	97	85	175	114	166	202	164	184	134	107	101	114
	PrecMin	599	4	2	5	11	24	58	56	26	37	20	16	15
	PrecStd	103	23	20	35	31	40	30	33	41	28	27	25	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.6	34.1	32.1	27.3	21.3	16.7	15.8	19.1	22.8	27.7	29.8	32.5
		MCWB	19.2	19.8	17.7	16.1	14.2	12.5	11.5	12.5	14.8	15.8	17.1	18.2
	2%	DB	30.0	29.2	27.6	23.2	18.5	15.0	14.2	16.5	19.7	23.2	25.7	26.8
		MCWB	18.2	18.5	17.1	14.9	13.3	11.9	11.0	11.3	13.2	14.9	17.4	17.1
	5%	DB	25.6	25.5	23.6	20.3	16.7	14.1	13.4	14.8	17.2	20.3	22.2	22.9
		MCWB	17.8	17.8	16.6	14.6	13.1	11.5	10.5	11.0	12.5	14.1	16.4	16.6
	10%	DB	22.0	22.2	20.6	17.9	15.5	13.2	12.6	13.4	15.3	17.4	19.1	19.9
		MCWB	17.1	17.3	15.9	14.1	12.7	11.0	10.1	10.5	11.6	13.2	15.0	15.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.4	21.6	19.6	17.6	15.5	13.7	13.0	13.7	16.0	17.4	19.8	19.7
		MCDB	27.6	27.3	26.6	22.9	17.9	15.1	14.4	16.6	19.6	23.7	25.4	26.6
	2%	WB	19.4	19.6	18.2	16.3	14.5	12.7	11.8	12.4	14.2	15.8	18.1	18.3
		MCDB	26.0	26.4	23.9	20.1	16.8	14.2	13.3	14.9	18.2	21.4	23.3	24.3
	5%	WB	18.3	18.5	17.3	15.4	13.8	12.0	11.1	11.6	13.0	14.6	16.6	17.0
		MCDB	24.0	23.8	21.5	18.5	15.8	13.5	12.6	13.9	16.2	19.1	21.5	22.5
	10%	WB	17.3	17.6	16.3	14.6	13.1	11.3	10.5	10.9	12.1	13.4	15.3	15.9
		MCDB	21.9	21.4	19.9	17.4	15.0	12.8	12.1	13.1	14.8	16.8	18.7	19.7
Mean Daily Temperature Range	5% DB	MDBR	8.6	7.9	7.9	7.6	6.3	5.7	5.5	6.1	6.9	7.8	7.8	8.2
		MCDBR	15.7	14.6	14.4	12.4	8.6	7.1	6.4	8.7	11.0	14.1	14.6	14.7
		MCWBR	5.9	5.5	5.7	6.1	5.2	5.0	4.4	5.1	6.2	7.2	7.0	6.4
	5% WB	MCDBR	12.8	11.4	11.0	9.5	6.8	5.9	5.6	7.2	9.1	12.1	12.6	12.4
		MCWBR	5.7	5.2	5.2	5.6	4.8	4.6	4.2	5.0	5.8	7.0	6.7	6.1
Clear-Sky Solar Irradiance	taub	0.340	0.339	0.326	0.318	0.306	0.304	0.303	0.314	0.338	0.338	0.336	0.337	
	taud	2.548	2.559	2.597	2.601	2.614	2.601	2.607	2.571	2.486	2.495	2.529	2.545	
	Ebn at Noon	993	970	943	888	838	807	834	879	913	959	989	1000	
	Edh at Noon	107	102	90	78	67	63	66	79	99	107	109	109	
All-Sky Solar Radiation	RadAvg	7.09	6.04	4.74	3.32	2.23	1.77	1.94	2.80	4.05	5.40	6.33	7.03	
	RadStd	0.37	0.33	0.24	0.22	0.12	0.11	0.11	0.15	0.27	0.37	0.40	0.46	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.42	N/A	N/A	N/A	+0.45	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	-91	+128	+38

Nomenclature: See separate page